Neural Population Dynamics Reveal Internal Representations Underlying Adaptive Obstacle Avoidance in Reinforcement Learning

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August 9, 2026

Abstract

Understanding why a learned controller succeeds or fails under unexpected disturbances is as important as knowing whether it succeeds. Here we train a Proximal Policy Optimization (PPO) policy to navigate a planar reaching task while avoiding obstacles, then probe it with graded lateral force perturbations spanning seven conditions: one unperturbed control (P0) and three increasing leftward (L1–L3) and three increasing rightward (R1–R3) impulses. We treat the policy's hidden-layer activations as a population signal and apply the same analytical toolkit used in systems neuroscience — dimensionality reduction, state-space trajectory analysis, and linear decoding — to ask what internal signals track behaviour and distinguish successful from failed trials.

Behaviourally, the controller maintained 100% success on P0 and all mild perturbations; success fell to 90% under the largest leftward impulse (L3) and to 50% under the largest rightward impulse (R3), revealing a direction-dependent asymmetry in robustness. Kinematic adaptation was mediated by trajectory-level corrections rather than changes in peak speed. At the representational level, the first two principal components of the policy's hidden-layer activity accounted for 74.8% of population variance and organized into four discrete activity clusters. Lateral velocity profiles in latent space cleanly separated leftward from rightward perturbation conditions, and a linear classifier trained on single-timestep hidden activity successfully decoded the most extreme conditions (L3: 70%, R3: 62%)

while confusing mild conditions with stronger ones.

These results show that a PPO policy trained purely for task success spontaneously develops structured, low-dimensional internal representations that encode perturbation context in a way that is consistent with population coding principles observed in biological motor systems.

1 Introduction

Artificial neural-network controllers are now practical for real-time sensorimotor tasks [1], and Reinforcement Learning (RL) — in which a controller is shaped entirely by trial-and-error interaction with an environment rather than hand-tuned rules — has become a standard approach for generating closed-loop motor policies [2, 3, 4]. The appeal is clear: RL produces controllers that adapt to task demands without requiring an explicit model of the environment. The limitation is equally clear: good task performance tells us almost nothing about what internal signals the network uses or why it breaks down under novel conditions [5, 6, 7, 8].

Human motor neuroscience offers a useful reference point. During reaching, the nervous system continuously updates motor commands in response to unexpected mechanical disturbances, drawing on sensory feedback and internal forward models [9, 10, 11, 12, 13]. This flexibility is not confined to individual neurons: distributed activity across motor cortex supports systematic adjustments in movement as task demands change [12, 14, 15]. Crucially, population activity in motor cortex evolves along smooth, compact neural manifolds — low-dimensional subspaces that capture the essential dynamics of movement generation [16, 17, 18, 19]. Dimensionality reduction, neural decoding, and trajectory analysis of population activity have consequently become central tools for studying how biological circuits generate behaviour [20, 21, 22, 23, 24].

RL policies are also implemented as neural networks, yet relatively little is known about the internal representations that emerge during learning. Most RL studies report behavioural metrics — task success, cumulative reward, sample efficiency — while treating the learned policy as a black box [3]. Mechanistic interpretability approaches have begun to examine what networks encode [5, 6], but few have asked whether RL controllers exhibit organizing principles analogous to those in biological motor populations, especially under perturbation.

This gap matters practically. Behavioural performance alone cannot explain why a controller that performs well under normal conditions fails under distribution shift. Identifying latent signatures associated with adaptive success and failure would support failure diagnosis and contribute to more trustworthy autonomous systems [7, 8, 25].

In this study, we train a PPO controller on an obstacle-avoidance reaching task and then probe it with the same graded perturbation logic used in motor neuroscience experiments. We record hidden-layer activity during evaluation and analyse it using standard population-level tools. Our objective is concrete: identify which internal signals track kinematics, encode perturbation context, and separate successful from failed trials.

Our contributions are: (1) an analysis pipeline that treats RL hidden-layer activity as population data, enabling direct comparison with neuroscience tools; (2) a behavioural characterization of perturbation-driven performance degradation, including a direction-dependent robustness asymmetry; and (3) evidence that low-dimensional latent structure encodes perturbation context in a linearly readable way, with extreme conditions producing the most distinct representations.

2 Methodology

2.1 Task Environment

We implemented the task using the Python Gymnasium interface [26], which provides a standard way to define observations, actions, rewards, and episode resets for physics-based simulations. The agent is a two-dimensional point mass that receives continuous force commands at each timestep and evolves under simple Newtonian dynamics within a bounded workspace. We deliberately keep the dynamics minimal so that any internal structure we find in the network cannot be attributed to complex physical constraints. For reader intuition, Figure 1A depicts the task conceptually using a planar arm analogy drawn from human motor neuroscience; all experiments use the point-mass dynamics.

The main experiment used a two-obstacle corridor (Figure 1C) that constrains the agent to pass between two static circular obstacles on its way from a fixed start to a fixed goal. A baseline single-obstacle experiment was used to pretrain the obstacle-avoidance skill before the perturbation protocol was introduced.

The agent observes its position and velocity together with the positions of the goal and both obstacles, all expressed relative to the agent's current location so the representation is invariant to where in the workspace the agent happens to be. Formally, the state at timestep t is:

$$s_t = \left[g_x - x_t,\ g_y - y_t,\ o_{Lx} - x_t,\ o_{Ly} - y_t,\ o_{Rx} - x_t,\ o_{Ry} - y_t,\ v_{x,t},\ v_{y,t}\right] \quad (1)$$

where (x_t, y_t) and $(v_{x,t}, v_{y,t})$ are the agent's position and velocity, (g_x, g_y) is the goal, and (o_L, o_R) are the left and right obstacle centres.

2.2 Reward and Perturbation Protocol

At each timestep the agent receives a scalar reward that combines four terms:

$$R_t = w_p \, \Delta d_{\text{goal}} - w_v \, \|v_t\| - w_c \, C_t + w_s \, S_t \tag{2}$$

Progress toward the goal is rewarded (Δd_{goal}); excessive speed is penalized ($\|v_t\|$); collision with an obstacle incurs a large negative penalty (C_t); and reaching the goal earns a terminal bonus (S_t).

To probe online adaptation, we applied a brief lateral force impulse mid-movement on a random subset of trials. The impulse was held constant for six consecutive timesteps:

$$f_x(t) = p_i, \quad t \in [50, 55] \tag{3}$$

where $p_i \in \{\pm 0.15, \ \pm 0.30, \ \pm 0.45\}$ sets the magnitude for episode i. This defines seven evaluation conditions: one unperturbed control (P0) and six perturbation levels — leftward L1 (-0.15), L2 (-0.30), L3 (-0.45) and rightward R1 ($+0.15$), R2 ($+0.30$), R3 ($+0.45$). The protocol mirrors perturbation designs used in biological reaching studies to investigate online feedback correction [9, 11, 12, 27].

2.3 Reinforcement Learning Controller

All training and evaluation used Stable-Baselines3 [28] with Proximal Policy Optimization (PPO) [3, 29]. PPO updates the policy by maximising a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \Big[\min\big(r_t(\theta) \hat{A}_t, \ \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \big) \Big] \tag{4}$$

where $r_t(\theta)$ is the ratio of the updated policy probability to the old policy probability at timestep t, $\hat{A}_t$ is the estimated advantage (how much better the chosen action was than average), and $\epsilon = 0.2$ is the clipping threshold that prevents the policy from changing too abruptly in any single update.

The controller uses separate policy and value networks, each a feedforward multilayer perceptron (MLP) with two hidden layers of 256 units. A feedforward MLP applies successive nonlinear transformations to its input at each timestep independently — it has no recurrent connections and no explicit memory of previous timesteps. Continuous actions are sampled from a diagonal Gaussian distribution whose mean is output by the actor. Observations were normalised online using a running mean and variance (VecNormalize) to improve training stability.

Training ran for 2×10^6 environment timesteps with learning rate 3×10^{-4}, discount factor $\gamma = 0.99$, entropy coefficient 0.005, value-function loss coefficient 0.5, and maximum gradient norm 0.5. All experiments were run with Stable-Baselines3 v2.8.0, Gymnasium, and PyTorch on an Intel® Core™ Ultra 7 255H processor.

2.4 Neural Population Analysis

After training, we extracted the hidden-layer activations of the policy network across all evaluation episodes and treated them as a time-resolved neural population recording. Let $h_t \in \mathbb{R}^{256}$ denote the activation vector of the first hidden layer at timestep t. The full population record for one episode of length T is:

$$H = \{h_1, h_2, \ldots, h_T\} \tag{5}$$

We applied three analyses to H. **Principal Component Analysis (PCA)** projects the high-dimensional activity onto a small number of dimensions that capture the most variance, revealing the geometry of the population state space. **Tuning analysis** regresses individual unit activity against instantaneous movement speed to ask whether specific units are selective for kinematic variables. **Linear decoding** trains a linear classifier on single-timestep hidden activity to test whether perturbation condition can be read out directly from the population state without nonlinear processing.

Behavioural metrics (episode reward, path length, duration, mean and peak speed, peak lateral velocity) were computed per episode and summarised with descriptive statistics. All analyses were implemented in Python (NumPy, SciPy, scikit-learn).

3 Results

We evaluated the trained PPO controller across all seven perturbation conditions and report results in four parts: (i) how trajectory geometry changed with perturbation magnitude and direction; (ii) the corresponding kinematic signatures; (iii) the low-dimensional structure of the policy's hidden-layer activity; and (iv) how well perturbation context can be decoded from that activity.

3.1 Behavioural Adaptation Under External Perturbations

Figure 2 shows reach trajectories overlaid across all seven conditions. Under the unperturbed control (P0) and all mild perturbations (L1, L2, R1), the controller reached the goal on every trial, generating smooth paths that curved symmetrically around the two obstacles. As perturbation magnitude increased, trajectories deflected proportionally in the direction of the applied force: leftward conditions (L1–L3) shifted paths to the left and rightward conditions (R1–R3) shifted them to the right.

Goal success degraded progressively with perturbation strength. Failures occurred only under the two largest rightward conditions (R2: 85%, R3: 50%) and under L3 (90%). The asymmetry between leftward and rightward conditions at the same nominal magnitude is visible in the trajectories: under L3 the path curves sufficiently to pass between the obstacles, whereas under R3 many paths collide with the right obstacle before the controller can correct. Across successful trials, peak movement speed was effectively constant across conditions, indicating that adaptation was achieved through changes in path geometry rather than by moving faster.

3.2 Kinematic Signatures of Perturbation Direction

Figure 3 characterises how lateral velocity evolved across conditions.

Panel A shows that lateral velocity profiles separated cleanly by perturbation direction: leftward conditions (L1–L3) remained below zero while rightward conditions (R1–R3) were above zero throughout the perturbation window. The magnitude of the deflection scaled with perturbation strength, and conditions on the same side of zero overlapped in the post-perturbation phase, suggesting that the controller applies a corrective force of proportional size rather than a discrete binary response.

Panel B confirms that total movement speed (Euclidean norm of velocity) increased rapidly at movement onset and plateaued at approximately the same level across all conditions. This pattern shows that the controller does not compensate for stronger perturbations by moving faster; instead it redirects the trajectory laterally while preserving the overall tempo of the movement, a strategy analogous to feedback-driven path correction in human reaching [11, 12, 30].

3.3 Low-Dimensional Structure of the Latent Neural State Space

To examine whether the policy's hidden-layer activity has internal structure, we projected the full population activity (256-dimensional) onto its first two principal components across all evaluation timesteps (Figure 4).

Two findings stand out. First, the activity does not fill 256-dimensional space uniformly: the first two principal components alone account for 74.8% of the total variance, indicating a compact, low-dimensional representation. This is consistent with neural manifold structure observed in motor cortex, where coordinated population activity during reaching can be captured in a small number of dimensions despite the large number of recorded neurons [16, 17, 18, 19].

Second, the manifold naturally subdivides into four clusters (Panel B). Because these clusters are not aligned with individual perturbation conditions, they likely reflect movement phase — for example, the acceleration phase near movement onset, the perturbation window, the correction phase, and the approach to goal — that is shared across conditions. Condition-specific differences appear as shifts in the density and spread of points within this shared geometry, rather than as completely separate manifolds. This is consistent with the idea that the controller uses a common computational substrate for obstacle avoidance, modulated by perturbation context rather than replaced by it.

3.4 Linear Decoding of Perturbation Context from Latent Activity

Finally, we asked whether perturbation condition could be read out from the hidden-layer activity using a simple linear classifier — that is, without any nonlinear processing of the population signal (Figure 5).

Panels A and B confirm that the policy converged stably during training and maintained consistent performance across evaluation checkpoints, providing confidence that the representations we analyse reflect a converged policy rather than a partially trained one.

The confusion matrix (Panel C) shows that a linear decoder trained on single-timestep hidden activity can classify perturbation conditions above chance, but performance is not uniform across conditions. Extreme perturbations produce the most distinct latent states: L3 was correctly classified 70% of the time and R3 62% of the time. Moderate rightward perturbations were also reasonably decodable (R1: 53%, R2: 50%). In contrast, mild and control conditions (P0, L1, L2) were frequently misclassified as stronger perturbation conditions — particularly L3 — suggesting that these states occupy a region of the manifold that is geometrically closer to the extreme left perturbation than to the control.

The fact that a linear decoder succeeds at all is informative: it implies that perturbation context is encoded in the population activity in a relatively direct, structured way, rather than being buried in nonlinear interactions among units.

This is consistent with the notion that the hidden activity does not merely store the immediate control command but also reflects the broader task context in which that command is generated.

4 Discussion

This study asked whether a PPO controller trained for adaptive obstacle avoidance develops internal representations that are interpretable in the same terms used to study biological motor populations. The short answer is yes, but with important qualifications.

Behavioural adaptation parallels biological reaching. The controller maintained robust goal-directed behaviour across mild perturbations and degraded progressively under stronger ones. Crucially, degradation was asymmetric: rightward perturbations were more disruptive than leftward ones of the same nominal magnitude, a pattern that likely reflects the geometry of the two-obstacle corridor and the learned bias toward the left-side route (see Figure 2). Across all conditions, adaptation was expressed through lateral trajectory corrections rather than changes in movement speed — a strategy that closely resembles the feedback-mediated path corrections observed in human reaching under mechanical perturbations [11, 12, 9, 30, 31]. We are cautious about overinterpreting this resemblance: the policy was optimized for reward, not to replicate biological motor strategies, and the point-mass dynamics are far simpler than a real limb. Nevertheless, the kinematic signature — proportional lateral deflection, preserved speed profile — is consistent with feedback correction principles in biological reaching.

Low-dimensional latent structure. The most striking representational finding is that 74.8% of population variance in a 256-unit network can be captured in just two dimensions. This degree of compression is consistent with neural manifold structure reported in motor cortex during reaching, where coordinated population dynamics unfold on low-dimensional subspaces despite the large number of neurons involved [16, 17, 18, 19, 20, 21]. The four-cluster structure within the manifold aligns with an intuitive decomposition of the task into movement phases (approach, perturbation, correction, arrival), suggesting that the network has learned to use qualitatively different activity regimes for qualitatively different moments in the trial. We note that this clustering was identified post hoc with K-means; future work with held-out data and multiple seeds would be needed to establish whether the four-cluster solution is stable across training runs.

What the decoder tells us. The linear decoder reliably identified extreme perturbation conditions (L3: 70%, R3: 62%) but struggled with mild ones, which were often classified as the large leftward perturbation. This pattern is not simply a failure of the decoder: it reflects the geometry of the latent space. Mild perturbations produce hidden states that are geometrically close to the extreme left perturbation state, suggesting that the network does not maintain sharply distinct representations for conditions that produce similar behavioural outcomes. This has a practical implication for interpretability: representational overlap between conditions means that the network's internal state alone is not always a reliable indicator of which perturbation is acting. The most distinctive signal comes from conditions that also produce the most distinctive behaviour (i.e., largest trajectory deviations or failures).

Limitations. Several constraints bound the scope of these conclusions. First, results are from a single training seed; we cannot quantify how much the representational structure varies across runs. Multiple independent seeds are needed to establish reliability. Second, the environment is highly simplified — a two-dimensional point mass with static obstacles and a fixed perturbation window. Whether similar representational organization emerges in environments with dynamic obstacles, contact-rich dynamics, or more complex sensory observations remains an open question. Third, the feedforward MLP has no memory of previous timesteps, which limits the controller's ability to maintain an internal estimate of perturbation history. Recurrent or transformer-based policies may exhibit richer temporal dynamics. Fourth, the linear decoding and PCA analyses reported here are descriptive; they confirm that structure exists and characterize its broad geometry, but do not provide a causal account of how specific units or circuits contribute to behaviour. Finally, large language models were used to support debugging, proofreading, grammar, and LaTeX formatting, and were not used to generate experimental data, run analyses, or make scientific claims.

Broader significance. The framework presented here is straightforward to apply: train a policy, extract hidden activity during evaluation, apply population-level tools. The value is not that RL controllers replicate biological motor circuits — they do not — but that population-style analyses can reveal whether internal representations are structured and task-relevant in a way that behavioural metrics alone cannot. This provides an additional diagnostic layer for understanding when and why adaptive policies work, which is particularly relevant for systems expected to operate under distribution shift.

5 Conclusion

We trained a PPO policy on an obstacle-avoidance reaching task and probed its internal representations using the same analytical tools applied to motor cortex recordings in human neuroscience. The controller adapted to graded lateral perturbations through trajectory-level corrections that preserved peak movement speed, with robustness degrading asymmetrically for the largest rightward impulses. The policy's hidden-layer activity was organized into a compact low-dimensional manifold accounting for 74.8% of population variance, subdivided into four activity clusters consistent with movement phases. A linear classifier trained on single-timestep hidden activity decoded extreme perturbation conditions above chance, while mild conditions were geometrically indistinct from stronger ones.

Together, these findings demonstrate that a purely reward-driven training process can produce internal representations that are structured, compact, and partially readable with simple linear tools. Treating policy hidden activity as population data provides an analytical lens that complements behavioural evaluation and may help explain why adaptive controllers succeed or fail under conditions they were not explicitly trained on.

Future work will extend this framework to richer environments, multiple training seeds, and recurrent architectures, and will examine whether representational structure generalises across task variants or degrades in ways that predict behavioural failure.

Acknowledgments

This work was undertaken thanks in part to funding from the Connected Minds Program, supported by Canada First Research Excellence Fund, Grant No. CFREF-2022-00010.

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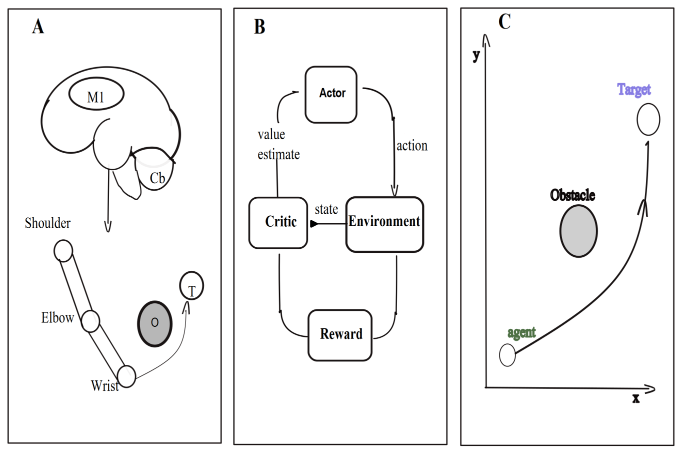

Figure 1: **Task, control architecture, and analysis pipeline.** **(A)** Conceptual analogy: a biological controller (primary motor cortex M1 and cerebellum Cb) drives a planar arm toward a target (T) while avoiding an obstacle (O). The arm is used for reader intuition only; all experiments use a two-dimensional point-mass agent. **(B)** The PPO actor–critic loop. At each timestep the *actor* selects a continuous force command from the current state observation; the *environment* applies that command, advances the agent, and returns the next state; the *critic* estimates the long-term value of each state to guide policy improvement; a scalar *reward* signal drives both networks. **(C)** Task geometry. The agent moves from a fixed start position (bottom) through a narrow corridor formed by two circular obstacles toward the goal region (top). Trajectories recorded during evaluation are aligned to this geometry for the population analyses described below.

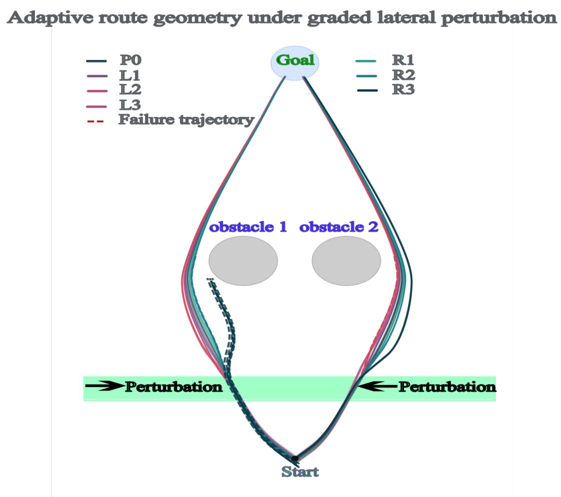

Figure 2: **Cumulative reach trajectories across all seven perturbation conditions.** All evaluation episodes are overlaid for each condition. Solid lines are successful trials; the dashed red line is a failure trajectory. Leftward perturbations (L1–L3, pink/mauve) deflect paths toward the left obstacle; rightward perturbations (R1–R3, teal/dark green) deflect them toward the right obstacle. The shaded green horizontal band marks the segment of the workspace in which the lateral force impulse is active. Two grey ellipses represent the static obstacles forming the corridor. Success rates by condition: P0 100%, L1 100%, L2 100%, L3 90%, R1 100%, R2 85%, R3 50%.

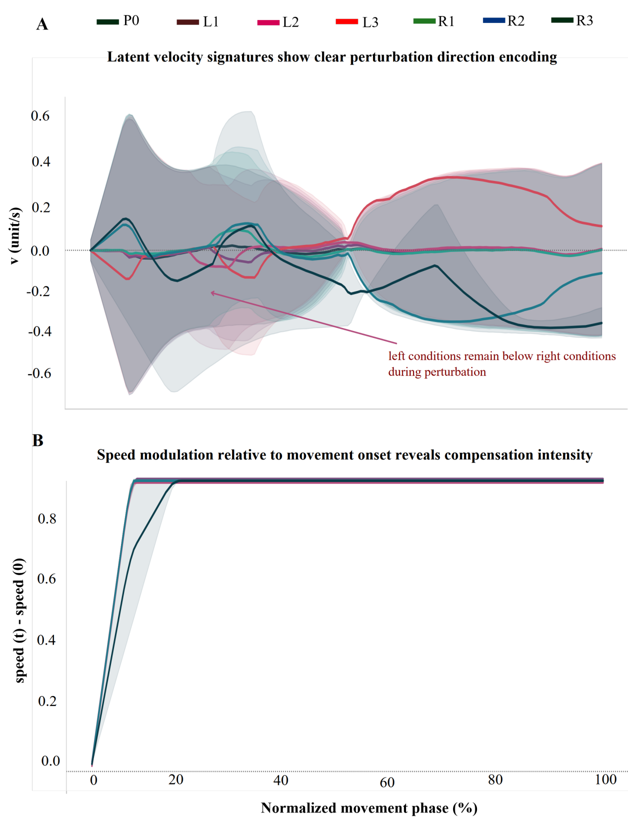

Figure 3: **Velocity profiles across perturbation conditions.** **(A)** Mean lateral velocity over time, averaged across episodes within each condition (shaded bands: ±1 SD). Leftward conditions produce negative lateral velocity (downward deflection); rightward conditions produce positive lateral velocity. The annotation marks the period during which left and right conditions diverge most clearly, corresponding to the perturbation window (timesteps 50–55). **(B)** Speed modulation relative to movement onset (normalised movement phase). Speed increases rapidly in the first 20% of the movement and then plateaus at approximately the same level across all conditions, confirming that perturbation adaptation is expressed through lateral redirection rather than overall speed change.

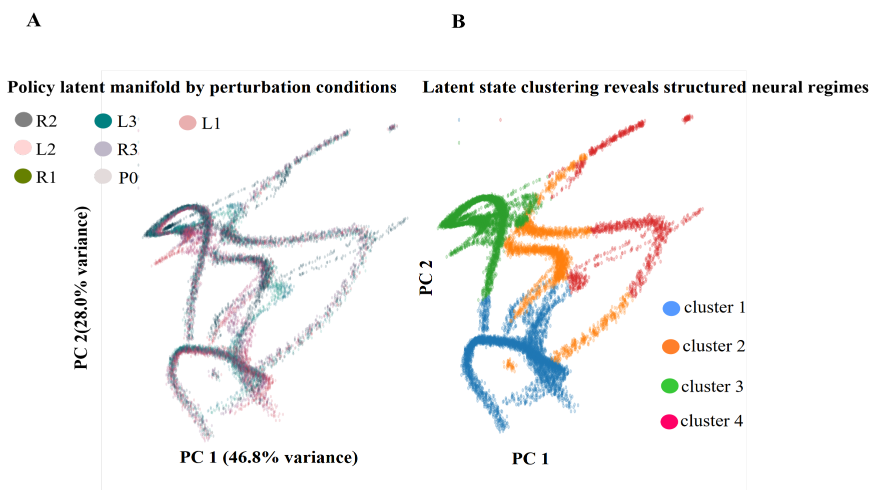

Figure 4: **Low-dimensional structure of the policy's hidden-layer activity.** Each point is the 256-dimensional hidden-state vector at one evaluation timestep, projected onto the first two principal components. **(A)** Points coloured by perturbation condition. The manifold has a consistent curved shape across all conditions, but individual conditions occupy partially separable regions, particularly at the extremes of the distribution. PC1 captures 46.8% of variance; PC2 captures 28.0%; together they account for 74.8%. **(B)** The same projection coloured by K-means cluster label ($K = 4$), showing that the manifold divides naturally into four distinct activity regimes. These clusters are not defined by condition — they likely correspond to phases of the movement (approach, perturbation, recovery, arrival) that are shared across conditions.

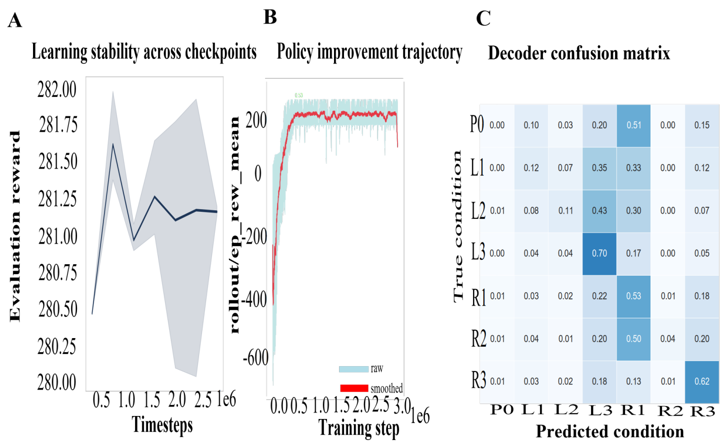

Figure 5: **Training dynamics and linear decoding of perturbation context.** **(A)** Evaluation reward across training checkpoints (mean ± SD across episodes), showing stable performance after convergence. **(B)** Training curve: rollout episode reward over 3×10^6 training steps (raw in light blue, exponential moving average in red). The policy converges from large negative initial rewards to a stable positive plateau within the first 1×10^6 steps. **(C)** Confusion matrix from a linear decoder trained to classify which of the seven perturbation conditions a given timestep belongs to, using only the 256-dimensional hidden-state vector at that timestep. Rows are true conditions; columns are predicted conditions; diagonal entries (darker blue) indicate correct classification. The decoder classifies extreme conditions most reliably (L3: 70%, R3: 62%, R1: 53%), while mild and control conditions (P0, L1, L2) are frequently misclassified as stronger perturbations, suggesting that mild perturbation states are geometrically closer to the stronger ones than to the control.